MIL-P-46522 (Ord)
27 April 1961

MILITARY SPECIFICATION

PROPELLANT, M17, FOR USE IN
CARTRIDGES, 76MM, AP-T, M339 AND
76MM, TP-T, M340A1

1. SCOPE

1.1 This specification covers one type of propellant designated as M17, for use with cartridges, 76MM, AP-T M339 and 76MM, TP-T, M340A1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

JAN-G-155	-	Graphite (For use in Ammunition)
JAN-N-244	-	Nitrocellulose (For use in Explosives)
MIL-N-246	-	Nitroglycerin
JAN-E-255	-	Ethyl Centralite (Carbonite)
JAN-N-494	-	Nitroguanidine (Picrite)
MIL-G-2550	-	General Specification for Ammunition Except Small Arms Ammunition

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109	-	Inspection Terms and Definitions.
MIL-STD-286	-	Propellants: Sampling, Inspection and Testing.

FSC 13752

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DRAWINGS

ORDNANCE CORPS

20-4-617	-	Marking Diagram and Sealing of Metal Lined Wooden Packing Boxes for Shipment of Propellant.
76-4-46	-	Box, Packing, with Metal Liner, M24 for Smokeless Powder. Assembly and Details.
76-4-56	-	Box, Packing, Metal-Wood, M17 for Smokeless Powder.

PUBLICATIONS

ORDNANCE CORPS

ORD-M608-11	-	Procedures and Tables for Continuous Sampling by Attributes.
IEL-(see 6.5)	-	Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-The constituent materials shall comply with the requirements of the applicable specification as follows:

Constituent Materials	Conforming to Specifications
Nitrocellulose	JAN-N-244, Grade C, Type I
Nitroglycerin	MIL-N-246, Type I
Nitroguanidine	JAN-N-494 (Class to be specified in contract) (see 6.1)
Ethyl Centralite	JAN-E-255, Class 2
Cryolite	Commercial grade of "Technical" sodium aluminum fluoride
Graphite	JAN-G-155

3.2 Composition.-The composition, exclusive of total volatiles, shall be as specified in Table I when determined as specified in 4.3.1.

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Table I Composition

Ingredient	Percent
Nitrocellulose	22.0 ± 0.75
Nitroglycerin	21.5 ± 0.75
Nitroguanidine	54.7 ± 1.0
Ethyl Centralite	1.5 ± 0.1
Cryolite	0.3 ± 0.1
Graphite	0.1 maximum (max.)

3.3 Total Volatiles.-The total volatiles content of the propellant shall be 0.30 percent max., when determined as specified in 4.3.2.

3.4 Heat test, 120 degrees Centigrade (°C.).-The propellant shall not cause complete fading of the methyl violet test paper to a salmon-pink color in less than 40 minutes, when tested as specified in 4.3.3.

3.5 Form and dimensions.

3.5.1 Form.-The propellant shall be in the form of cylindrical multiperforated grains.

3.5.2 Dimensions.-The following requirements regarding grain dimensions shall govern when determined as specified in 4.2.3.2, unless authorization for departure is given by the procuring activity concerned prior to manufacture of the propellant.

3.5.2.1 Length - diameter ratio.-The average grain length (L) shall be from 2.10 to 2.50 times the average grain diameter (D).

3.5.2.2 Length - diameter uniformity.-The length and diameter of the grain shall comply with the uniformity requirements shown in Table II.

Table II

Standard Deviation of Individual Dimensions Expressed as Percentage of Mean Dimension

Dimension	Max.
L	6.25
D (grains 0.2 inch or more in diameter)	3.125

3.5.2.3 Grain diameter - perforation diameter ratio.-The average grain diameter (D) shall be not less than 5 nor more than 15 times the average diameter of the perforations (d).

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3.5.2.4 Web measurements.-The difference between the average outer web thickness (W_o) and the average inner web thickness (W_i) shall be not more than 15 percent of the average thickness (W_a).

3.6 Ballistics.-The propellant, when fired in the weapon and with the weight of projectile for which it was manufactured, shall comply with the following requirements when tested as specified in 4.3.4.

3.6.1 Velocity level and pressure.-The propellant, at 70 degrees Fahrenheit (°F.) shall be assessed to impart a service velocity of 3,200 feet per second (f.p.s.) to the projectile without exceeding a rated max. pressure of 46,000 pounds per square inch (p.s.i.).

3.6.2 Uniformity of velocity.-The standard deviation in velocity for the lot at 70°F. shall be not larger than 6.5 f.p.s. (see 6.3).

3.6.3 Low temperature.-When the propellant is conditioned at minus 40°F. and fired at the assessed charge, the pressure shall not exceed the average pressure at 70°F.

3.7 Workmanship.-The best commercial practices shall be used in the formulation of propellant furnished under this specification, and all other applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes. The propellant shall not contain excessive quantities of cracked, distorted, short or long and otherwise deformed grains.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provision.-The supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein. Inspection shall be performed in accordance with this specification and other specifications referenced in any of the contractual documents.

4.1.1 Contractor quality assurance system.-The contractor shall provide and maintain an effective quality assurance system acceptable to the Government covering the supplies procured under this specification and shall perform all inspections required by the system. A current written description of the system shall be submitted to the contracting officer, and meet his approval, prior to initiation of production,. It shall include a description covering controls for lot formation and identification, inspection, inspections to be performed, inspection stations, sampling procedures, methods of inspection (gages, measuring and testing equipment), and provisions for control and disposition of non-conforming material. The written description will be considered acceptable when, as a minimum, it provides the quality assurance required by the provisions of 4.2 and 4.3 (except proving ground tests), and the other documents referenced herein. The contractor shall not be restricted to the inspection station nor the method of inspection listed in this specification provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether certain procedures of the contractor's system provide equal assurance, the comparable procedure of this specification shall apply. The contractor shall notify the Government of, and obtain approval for, any change to the written procedure that affects the degree of assurance required by this specification or other documents referenced herein.

4.1.2 Submission of product.-At the time the completed lot of product is submitted to the

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Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

b. Number of units of product inspected.

c. Results obtained for all inspection performed

d. Drawing, specification number and date, together with an identification and date of changes.

e. Certificates of analyses on all material procured directly by the contractor, when such material is covered by referenced Government specifications.

f. Number of items in the lot.

g. Date submitted.

h. Eight copies of Propellant Description Sheet, OO Form-1204, containing the applicable data.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification shall consist of: (a) surveillance of contractor operations to determine that the methods and procedures of the written procedure are being properly applied and (b) Government product inspection. The Government inspector shall also ascertain prior to acceptance that the product complies with all the quality assurance provisions of other specifications referenced in any of the contractual documents.

4.1.3.1 Surveillance.-Surveillance shall include, but is not limited to:

a. Observation of procedures concerning lot formation and identification.

b. Observation of sampling procedures and application of acceptance criteria.

c. Determination that all required examinations and tests are performed in accordance with the prescribed procedures of this specification, or approved equivalents thereto.

d. Review of procedures for control and disposition of non-conforming material.

4.1.3.2 Product inspection.-Physical inspection by the Government of contractor accepted product will be performed to the extent necessary to assure contractor compliance with the requirements of this specification and to determine the reliability of contractor records. Severity of Government inspection of individual characteristics will be directly related to the seriousness classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records. Specification MIL-G-2550 will apply.

4.1.3.3 Deviations and deficiencies.-Deviations from the prescribed or agreed upon procedures, or instances of poor practice which might have an effect on the quality of the product, shall be called to the attention of the contractor immediately. To avoid interference with operations, the contractor shall designate a responsible official, or officials, to whom the Government inspector will report such instances. Failure of the contractor to correct deficiencies

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promptly shall be cause for suspension of acceptance until correction has been made or until conformance to prescribed criteria has been demonstrated.

4.2 Inspection provision.

4.2.1 Lot formation.-A lot shall consist of one or more batches produced by one manufacturer, in accordance with the same specification or same specification revision, under one continuous set of operating conditions. Each batch shall consist of that quantity that has been subjected to the same unit chemical or physical process intended to make the final product homogeneous.

4.2.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105. Continuous sampling plans in accordance with Handbook ORD-M-608-11 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Box or container, prior to filling (see drawing (dwg.) 76-4-46, 76-4-56).

Catergories	Defects	Method of inspection
Critical: None defined		
Major:	AQL 0.25 percent	
101.	Foreign matter or rust on interior	Visual
Minor: None defined		

4.2.2.2 Sealed box or container (see dwg. 76-4-46, 76-4-56).

Categories	Defects	Method of inspection
Critical: None defined		
Major:	AQL 0.40 percent	
101.	Cover improperly closed	Visual-Manual
102.	Gasket missing or broken	Visual
103.	Wood cover split or otherwise damaged	Visual
Minor:	AQL 2.5 percent	
201.	Car seal missing, unsealed or improperly positioned	Visual-Manual
202.	Crack or hole in metal container	Visual
203.	Hardware improperly engaged	Visual
204.	Marking misleading or unidentifiable	Visual
205.	Bare areas on exterior coating of metal container the sum of which is in excess of 1/4 square inch	Visual

4.2.3 Testing.

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4.2.3.1 Chemical composition, stability and surveillance.-Ten pounds of propellant shall be selected at random from the lot. Five pounds shall be used for chemical composition and stability tests to be performed at a Government approved laboratory (see 6.6). The remaining five pounds shall be forwarded to the Commanding Officer, Picatinny Arsenal, Dover, New Jersey for 65.5°C. surveillance tests. All samples shall be packed in air-tight containers and shall be marked to show the propellant designation, the projectile for which manufactured, lot number, manufacturer, date of sampling contract number and number of pounds in the lot.

4.2.3.2 Form and dimensions of grains (see 3.5) Major defect.-Thirty grains shall be selected from each lot for these examinations. If the sample fails to meet the requirements specified, the lot shall be rejected. Examination for form shall be visual; measurements and calculation for dimensions shall be conducted in accordance with Standard MIL-STD-286, Method 504.1.

4.2.3.3 Ballistics.-One hundred and fifteen pounds of propellant shall be selected at random from the lot - 110 pounds for proving ground test and 5 pounds for closed bomb test. The test shall be performed as specified in 4.3.4 (see 6.2.2).

4.2.3.3.1 Pressure, max - Critical defect.-If the mean of the max. pressure results for the test round at 70°F., when corrected to the corresponding pressure at service velocity, exceed 98 percent of the rated max. pressure or if the mean of max. pressure results for the minus 40°F. test exceeds the mean of the max. pressure obtained at 70°F. the lot shall be rejected (see 3.6.1 or 3.6.3).

4.2.3.3.2 Uniformity of velocity - Major defect.-If the product of the factor 0.73 times the standard deviation in velocity of the test rounds exceeds the applicable requirements, the lot shall be rejected (see 3.6.2).

4.2.3.3.3 Retest.-If for any reason the proof director considers that the test conditions have detrimentally affected the test results, he shall declare the results invalid and perform a new test on additional rounds as required.

4.2.3.4 Air pressure test (see 5.1.1.1).-The containers shall be checked 100 percent as specified in 4.3.5. If any container fails to meet the requirement it shall be rejected and removed from the lot.

4.2.4 Inspection equipment.-For the performance of the examinations and tests specified in 4.2 and 4.3, when applicable, the contractor shall have available, and utilize correctly, all necessary gages, measuring and testing equipment (see 6.5). When inspecting the contractor's inspection equipment the Government inspector will determine that the contractor has available, and utilizes correctly, gages, measuring and testing equipment of the required accuracy and precision and that the instruments are of the proper type and range to make measurements of the required accuracy. The contractor will have available a set of master gages, standards and appropriate instruments for regularly scheduled calibration of his inspection equipment. Records of such regularly scheduled calibration will be maintained by the contractor and made available for review by the Government. The calibration of gages, standards and instruments will be periodically checked by authorized Government personnel.

4.3 Test methods and procedures.

4.3.1 Compostion.

4.3.1.1 Nitroglycerin.-The percentage of nitroglycerin shall be determined in accordance with Standard MIL-STD-286, Method 208.1 using pentane-methylene chloride azeotrope (two volumes of technical grade pentane to one volume of methylene chloride) as the solvent for extraction except that the nitroglycerin content shall be calculated on a graphite (glaze) and total volatiles-free basis.

4.3.1.2 Nitroguanidine.-The percentage of nitroguanidine shall be determined in

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accordance with Standard MIL-STD-286, Method 213.1 using pentane-methylene chloride azeotrope as the solvent for extraction, except that the nitroguanidine content shall be calculated on a graphite (glaze) and total volatiles-free basis.

4.3.1.3 Cryolite (sodium aluminum fluoride).-The percentage of cryolite shall be determined in accordance with Standard MIL-STD-286, Method 307.1, except that the percentage of cryolite shall be calculated on a graphite (glaze) and total volatiles-free basis.

4.3.1.4 Graphite.-The percentage of graphite shall be determined in accordance with Standard MIL-STD-286, Method 308.1.

4.3.1.5 Nitrocellulose.-The percentage of nitrocellulose shall be determined in accordance with Standard MIL-STD-286, Method 209.2.

4.3.2 Total volatiles.-The percentage of total volatiles shall be determined in accordance with Standard MIL-STD-286, Method 103.3.

4.3.3 Heat test, 120°C.-The 120°C. heat test shall be conducted in accordance with Standard MIL-STD-286, Method 404.1.

4.3.4 Ballistics.-Unless otherwise specified herein, the test shall be performed in accordance with applicable provisions of the Ordnance Proof Manual. A weight of charge shall be calculated (from closed bomb tests or from charge velocity and charge-pressure curves obtained by 80, 90, 100 and 105 percent estimated charge firings using a total of 8 test rounds) to give the prescribed service velocity. A ten round uniformity series shall be fired alternately with ten rounds from a standard reference lot of propellant. The mean max. pressure, mean velocity and the velocity standard deviation for the test and reference rounds shall be separately calculated. If the corrected mean velocity of the test rounds differs by more than one percent from the prescribed service velocity, re-evaluations of the established charge and a new uniformity series shall be accomplished prior to determination of the final recommended charge weight. This procedure shall be conducted with rounds conditioned to a temperature of 70 $\pm$ 2°F. A separate uniformity series of 7 rounds shall also be fired at minus 40°F. $\pm$ 2°F. for evaluation of max. pressure. Unless otherwise stated the test shall be performed on the same day with the same gun tube and weight of charge as the test at 70°F.

4.3.5 Air pressure test for containers.

4.3.5.1 Two to three pound test (see 5.1.1.1).-The container shall be subjected to an internal pressure of 2 to 3 p.s.i. by a method satisfactory to the contracting officer's representative. A gage (designed to read 1/4 p.s.i.) or a water manometer shall be assembled in the system. A drop of 1/4 p.s.i. on the gage or 0.7 inches on the manometer in 15 seconds shall be cause for rejection.

4.3.5.2 One half to one pound test (see 5.1.1.1).-By a suitable means the container shall be subjected to an internal pressure of 1/2 to 1 p.s.i. A water manometer shall be assembled in the system. A drop of 0.7 inches on the manometer in 15 seconds shall be cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 Level A.-The propellant shall be packed in accordance with dwg. 76-4-46 or 76-4-56.

5.1.1.1 Air pressure test of container.-Immediately prior to packing, the containers shall be tested with compressed air, at an internal air pressure of 1/2 to 1 p.s.i. for containers conforming to dwg. 76-4-56, and at an internal air pressure of 2 to 3 p.s.i. for all other

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containers. The containers shall withstand the applicable specified pressure for 15 seconds minimum (min.) without leakage.

5.2 Marking.-The containers shall be marked in accordance with dwg. 20-4-617.

5.3 Data cards.-Data card information shall be as specifed in Specification MIL-G-2550.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Packing requirements (see 5.1.1).

c. Class of Nitroguanidine required (see 3.1).

d. Details of the manufacturing process and the equipment used by the contractor will be submitted to the procuring activity in writing prior to commencement of manufacture. Any departure from this manufacturing process must be submitted for approval, in writing, to the procuring activity prior to initiation.

6.2 Surveillance and closed bomb.

6.2.1 Surveillance.-Forward the sample to the Commanding Officer, Picatinny Arsenal, Dover, New Jersey, for the 65.5°C. surveillance tests. All samples shall be packed in air-tight containers and shall be marked to show the propellant designation, the projectile for which manufactured, lot number, manufacturer, date of sampling, contract number and number of pounds in the lot.

6.2.2 Closed bomb.-Forward the sample to the above address and pack and mark as specified in 6.2.1. This sample is for the determination of relative quickness and relative force. The firing record shall indicate the date of this shipment.

6.3 Standard deviation.-The following formula or its equivalent shall be used to compute the sample standard deviation:

$$S = \sqrt{\frac{\Sigma(\bar{x} - x)^2}{n - 1}}$$

Where:

S = sample standard deviation.

X = sample mean value

x = individual observed value

$\sum (X - \bar{X})^2$ = the sum of the squares of the differences between the sample mean value and each individual observed value.

n = sample size.

6.4 Submission of test data.-In addition to the normal distribution of records, when the propellant is procured by the Department of the Army; one copy of all ballistic test data will be forwarded to the following:

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Commanding Officer
Picatinny Arsenal
ATTN: ORDBB-ND4
Dover, New Jersey

6.5 Inspection equipment.-Inspection equipment designs were not available at the time this specification was prepared. Prior to procurement, the procuring agency should contact Picatinny Arsenal, ATTN: ORDBB-ND, Dover, New Jersey, to determine availability of inspection equipment designs.

6.6 Cost of expense of surveillance.-The cost will be borne by the Government.

Notice.-When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:
Army-Ordnance Corps

Preparing activity:
Army-Ordnance Corps

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